

STEMsters Robotics Competition (SRC)

The word "SUMMIT" is rendered in a large, bold, sans-serif font. The letters are a light blue color. The 'S' and 'U' are a slightly darker shade of blue than the 'M', 'M', 'I', and 'T'.

SUMMIT

2026 Game Manual

October 24, 2026
Arcadia, CA

Version 1.0 | September 14, 2026
STEMsters is a student-led 501(c)(3) nonprofit

Summit

- **The game.** Two autonomous robots, red and blue, on a small raised hill. Push the other robot off the Hill, or be higher on the Hill when time runs out.
- **The robot.** 10 cm by 10 cm at the start, 500 g or less, no height limit, controlled by a micro:bit. Every other part is up to the team.
- **The match.** Three minutes. The referee's field controller starts and stops both robots by radio.
- **The team.** One to four students and one adult who stays for the whole event.
- **The day.** Qualification matches, then a single-elimination bracket. The schedule and awards are on the event page.

Climb together. Compete hard, help other teams when they need it, respect the referees and volunteers, and handle a win and a loss the same way. This rule applies to students and adults.

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1. About SRC and Summit

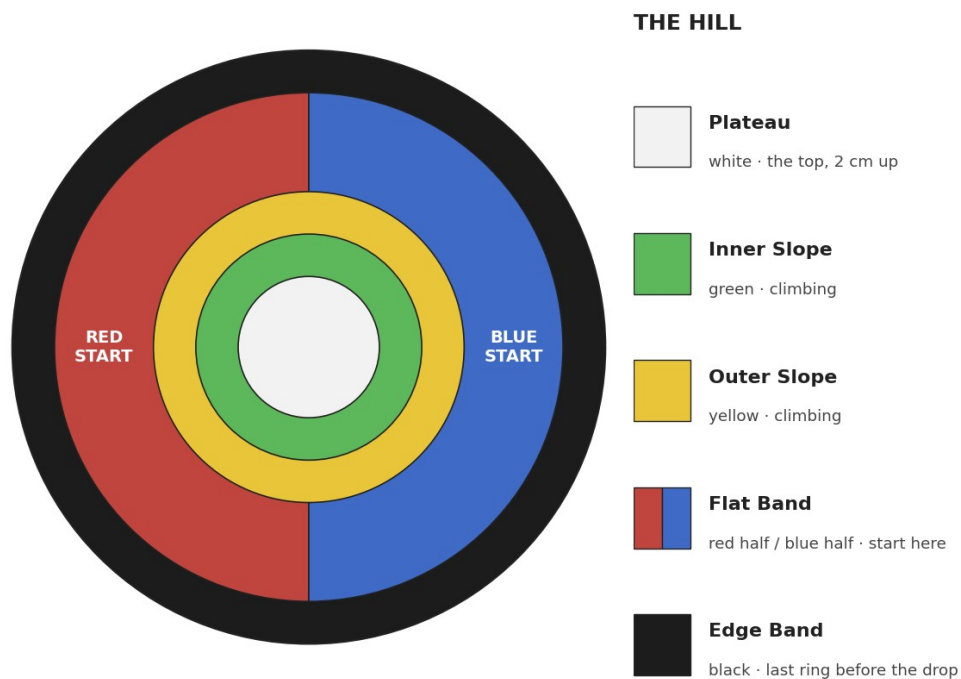
The STEMsters Robotics Competition (SRC) is an autonomous robotics competition for middle and high school students in Southern California. It is run by STEMsters, a student-led 501(c)(3) nonprofit. Summit is the SRC game for 2026.

This manual is updated in numbered versions. Each event runs on the version released before it. This is Version 1.0, for the October 24, 2026 event. The version and date appear on every page.

Summit is a sumo match between two autonomous robots on a small raised hill. A robot wins by pushing its opponent off the Hill, or by being higher on the Hill when the match ends. Robots run on their own code. Once a match starts, no one may touch or control them.

SRC is designed to be low-cost. The optional kit and registration together cost under \$150. Apart from the micro:bit, every part on the robot is the team's choice.

2. The Hill



Side view: the Plateau sits 2 cm above the Flat Band. The whole Hill is raised on a platform.



The Hill is a 3D-printed PLA mound, 42 cm across, raised on a platform. Each ring is a different color so a color sensor can tell where the robot is. Dimensions are in Appendix A. From the top:

- **Plateau (white).** The flat top, 2 cm above the base.
- **Inner Slope (green) and Outer Slope (yellow).** The two sloped rings between the Plateau and the base.
- **Flat Band (red and blue).** The flat base. The red robot starts on the red half and the blue robot on the blue half.
- **Edge Band (black).** The outermost ring. Its outer edge is the boundary.

3. The match

3.1 Before the match

1. When a match is called, one rostered student, the handler, brings the robot to the field. Only handlers and referees may be at the field.
2. The referee checks that the robot is powered on, has its team sticker, and fits its starting size.
3. **Red places first.** The schedule assigns colors. In qualification matches, colors are random. In the bracket, the lower seed is red.
4. The red handler places the robot anywhere fully on the red half of the Flat Band, facing any direction, and steps back. The blue handler then places on the blue half. A robot may not be touched after it is placed.
5. **Start.** The referee starts the field controller. It counts down five seconds and sends START to both robots at the same time. A robot that moves before START has false-started.
6. **Stop.** At 3:00 the field controller sends STOP.
7. **Backup.** If the field controller is not working, the referee counts down and both handlers press their robots' start buttons together. The robot's own five-second delay then applies.

3.2 End of the match

- **Ring-out.** Any part of a robot touches a surface other than the Hill: the platform, the table, or the floor. The match ends and the other robot wins. A robot hanging over the edge without touching anything is still in. If both robots go out and the referee cannot tell which touched first, the match is replayed.
- **Time.** At 3:00 the match ends. The robot in the higher zone wins (Section 4).
- **Dead stall.** If neither robot changes position for 30 seconds, the referee counts the last ten seconds aloud and ends the match. The robot in the higher zone wins.

3.3 During the match

- A robot that stops moving is not penalized. It scores wherever it is when the match ends.
- A flipped or stuck robot stays in the match and may right itself.
- If a part falls off, the match continues. A part that lands off the Hill is not a ring-out.
- The match is replayed if the field fails, someone interferes, or a referee error changes the result.
- A robot that smokes, sparks, or shorts is stopped immediately. It loses the match and must pass reinspection before it competes again.
- No one may touch a robot or the field, or send a signal to a robot, while a match is running.

4. Scoring

- **Result.** A match is a win, a loss, or a tie. A ring-out is a win. At time or dead stall, the robot in the higher zone wins. From the top, the zones are 1 Plateau, 2 Inner Slope, 3 Outer Slope, 4 Flat Band, 5 Edge Band.
- **Zone.** A robot's zone is the highest zone touched by any of its wheels or treads when the match ends. A robot with one wheel on the Plateau is in Zone 1. Arms, wedges, and skirts do not count.
- **Same zone.** The robot whose closest wheel is nearer the center of the Hill wins. The referee measures with a marked cord. If the difference is under 1 cm, the match is a tie.
- **Qualification points.** Win 3, tie 1, loss 0. Teams are ranked by points, then wins, then head-to-head, then random draw. A forfeit is a loss, and a win for the opponent.
- **Bracket.** Each pairing is a series. The first team to two wins advances. A tie counts for neither team and the series continues.

5. Robot rules

R1 Starting size. Robots must fit inside a 10.0 cm by 10.0 cm vertical footprint at the start of a match, checked with a gauge box. The box is cut a few millimeters oversize for build tolerance. A robot that fits the box passes. There is no height limit.

R2 Expansion. After START, robots may expand without limit. A robot must stay in one piece. Parts may not be detached, dropped, or launched.

R3 Weight. Robots must weigh 500.0 g or less in any configuration, battery included.

R4 Controller. A BBC micro:bit running the team's own code must control the robot's motion. All other parts are the team's choice. The official kit is optional.

R5 Radio. Robots may not be remote controlled. The only permitted wireless communication is receiving START and STOP from the field controller through the official start/stop code, which is part of the starter code. Robots may not transmit. Bluetooth must be off. Sensors, beacons, and markers may not be placed anywhere except on the robot.

R6 Manual start. Robots must also have a physical start button (the micro:bit's A or B button is acceptable) with a five-second delay before moving. It is used only when the field controller is not working.

R7 Power cutoff. Robots must have a switch or connector, reachable without tools, that cuts drive power immediately.

R8 Batteries. Any commercially made battery is allowed. There is no voltage limit. Batteries must be securely mounted and undamaged.

R9 Team number. Teams receive a team number by email at registration and enter it once in the official start/stop code. The robot shows the number on the micro:bit LEDs at power-on, and the field controller uses it to address the robot. At inspection, SRC applies a sticker with the same number. Keep the sticker visible. Nothing needs to be on the robot before the event.

R10 Entanglement. Nets, string, ribbon, and flexible parts intended to snare the other robot are not allowed. Belts, wiring, treads, and passive brush skirts are allowed.

R11 Wheels and surfaces. Adhesives, sticky or treated tires, suction, liquids, powders, and loose material are not allowed. Solid rubber, silicone, plastic, and printed wheels are allowed. At inspection, a sheet of plain paper is pressed onto each wheel and the robot is lifted. If the paper lifts with the robot, the wheel fails.

R12 Magnets. Magnets are not allowed on the robot except those inside motors and sensors. The Hill contains no steel. A magnet's only use is to grab the other robot or interfere with its compass.

R13 Weapons. Spinners, hammers, saws, blades, spikes, and sharpened edges are not allowed. Wedges, plows, scoops, lifters, and flippers are allowed.

R14 Sensor interference. Devices intended to jam, blind, or spoof the other robot's sensors are not allowed. A robot's own sensors may emit light and sound.

R15 Other prohibited items. Flame, heat, chemicals, projectiles, and glass.

R16 One robot. Each team competes with one robot. Spare parts are allowed. A second robot is not.

R17 Changes at the event. Teams may repair and reprogram between matches. After a major repair or code change, the robot must be reinspected before its next match.

6. Team rules

E1 Team size. A team is one to four students. A student may be on only one team.

E2 Roster. The names on the registration are the team's roster. Rosters are posted on the SRC website as first name and last initial. Only rostered students may work on the robot at the event. Rosters may be changed in the registration system until check-in on competition day.

E3 Level. Middle school is grades 6 through 8. High school is grades 9 through 12. Grades are for the 2026-27 school year. A team with students from both is a high school team. Level determines eligibility for the Middle School Champion and Excellence awards only. All teams compete together.

E4 Region. Summit is open to teams in Southern California. Teams from outside the region should contact SRC before registering.

E5 Adult. Each team must register a coach or supervising adult, 18 or older, who stays for the whole event, signs the waiver and the student-built pledge at check-in, and is responsible for the team. A team without its adult present may not compete. Independent teams are welcome under the same rule.

E6 Forms. Each student must have a completed consent and release form, available on the SRC website at [form link]. Students under 18 need a parent or guardian signature. Bring the form to check-in if it was not submitted online.

E7 Student-built. Students design, build, and program the robot. Adults may teach and advise. Adults may not build, repair, or program the robot.

E8 Registration. Registration opens September 14, 2026 on the SRC website at [registration link]. The fee is \$20 per team and covers the competition fields and awards. Team numbers are sent by email at registration. The deadline, kit ordering, and field ordering are on the same page.

7. Tournament

T1 Inspection. Every robot must pass inspection before its first match, using the checklist in Appendix C. Teams may retry as many times as needed before the inspection deadline. Students make the corrections. Inspectors explain but do not touch the robot. A robot that has not passed by the deadline forfeits its matches until it passes. Referees may reinspect any robot at any time.

T2 Format. Middle school and high school teams compete together: one set of qualification rankings and one bracket.

T3 Qualification. Every team plays the same number of matches against randomly assigned opponents. The number is set on the day from the number of teams checked in and is posted with the schedule.

T4 Bracket. Single elimination, seeded from qualification, with byes to the top seeds. Each pairing is first to two wins. The lower seed is red for the whole series.

T5 Match calls. Teams queue at their field before the match is called. A team not at its field within one minute of the call forfeits the match.

T6 Questioning a call. A rostered student may question a call at the field immediately after the match. The head referee's decision is final. Adults may not approach referees about calls.

The event date, schedule, calibration time, awards, and exhibition field are on the event page at [event page link].

8. Climb together: conduct and safety

Climb together is the SRC conduct rule. Compete hard, help other teams when they need it, respect the referees and volunteers, and handle a win and a loss the same way. Lending a part, helping another team debug, or cheering a good match is what SRC is for, and it is what the award judges ask about.

- Every robot restriction in Section 5 is a safety rule. It is checked at inspection and enforced all day.

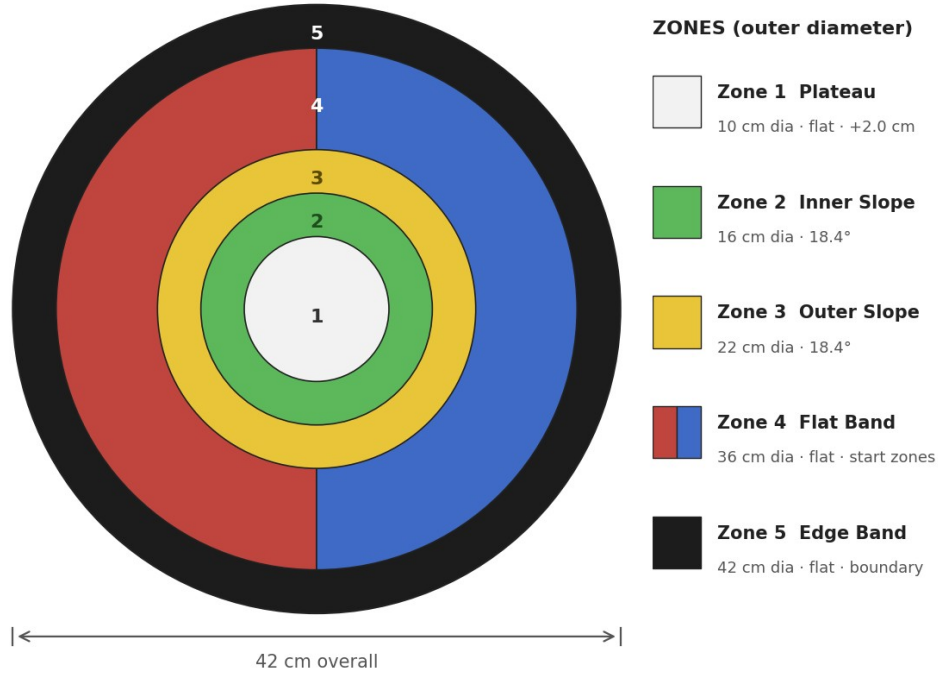
- Batteries must be secured. Charge only in the charging area and stay with the charger. Report damaged, swollen, or hot batteries to event staff immediately.
- Safety glasses are required when using tools. Hand tools only. No soldering irons, power tools, or anything that produces flame, sparks, or fumes.
- Treat opponents, volunteers, referees, and spectators with respect, in the stands, in the team area, and online. This applies to coaches and parents as well as students.
- The team's adult is responsible for the team's safety and conduct throughout the event. In an emergency, follow event staff.

9. Violations

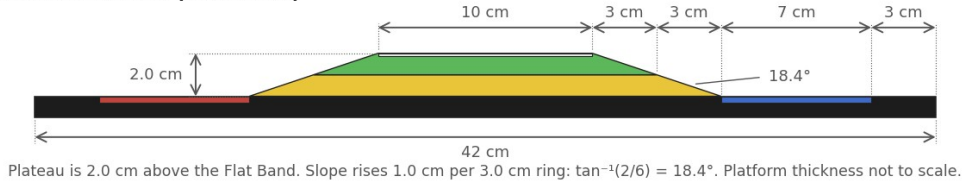
Violation	Consequence
A robot moves before START, or a handler touches a robot after placing it	First offense: both robots are re-placed and the match restarts. Second offense in the same match: that team loses the match.
Team not at the field within one minute of the call	Forfeit, recorded as a loss.
Robot has not passed inspection	May not compete. Matches called before it passes are forfeits.
Illegal item found on a robot after a match	The match is recorded as a loss. The robot must be corrected and reinspected before its next match.
Part detached on purpose	Loss of the match.
Remote control, a robot transmitting, or any signal sent to a robot during a match	Disqualification from the event.
An adult built, repaired, or programmed the robot	Disqualification from the event.
Robot smokes, sparks, or shorts	Match stopped and recorded as a loss. Reinspection required.
Intentional damage to the field or another robot	Loss of the match. Second offense: disqualification.
A student or adult violates the Climb together rule	Warning, then loss of the team's next match, then disqualification.
An adult approaches referees about a call	Warning to the team. Second offense: loss of the team's next match.

Appendix A. Field specification

TOP VIEW



CROSS-SECTION (true scale)



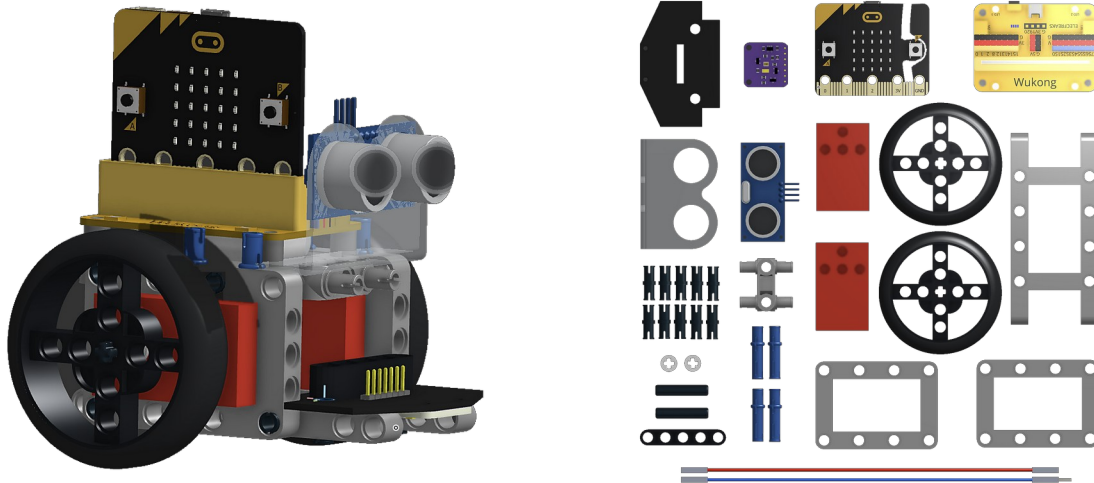
Zone	Inner radius (cm)	Outer radius (cm)	Surface	Color
1 Plateau	0	5.0	Flat, 2.0 cm above base	White
2 Inner Slope	5.0	8.0	18.4 degree slope, rises 1.0 cm	Green
3 Outer Slope	8.0	11.0	18.4 degree slope, rises 1.0 cm	Yellow
4 Flat Band	11.0	18.0	Flat, base level	Red half, blue half
5 Edge Band	18.0	21.0	Flat, base level	Black

- Each slope ring is 3.0 cm wide and rises 1.0 cm. The slope is $\arctan(1/3) = 18.4$ degrees across both rings.
- The mound is unpainted PLA on an MDF base. There are no lines or texture changes other than the color boundaries. The Hill sits on a raised platform with a catch area around it.
- The mound is printed in single-color wedges split along the ring boundaries.

- Practice fields, print files, the filament for each zone, and reference readings from the kit color sensor are on the SRC website at [event page link]. Readings vary with the sensor, its height, and the lighting. Treat them as a starting point and calibrate on a real field.

Appendix B. Official kit

The kit is optional. It is a complete, competition-legal robot. Teams may build it and compete with it as-is, or use it as a starting point. It is programmed in MakeCode.



- BBC micro:bit (sold separately)
- ELECFREAKS Wukong expansion board with built-in battery
- Two Geeservo 2 kg 360-degree continuous-rotation motors on M1 and M2 as the drive
- HC-SR04 ultrasonic sensor for finding the opponent
- TCS34725 color sensor for reading the zone
- LEGO Technic chassis parts
- Starter code, including the official field-controller start/stop code and the five-second manual-start delay

Kit ordering and price are on the SRC website at [event page link].

Appendix C. Inspection checklist

- Fits the gauge box in its starting configuration (R1).
- 500.0 g or less on the event scale in its heaviest configuration (R3).
- Motion is controlled by a micro:bit running the team's code (R4).
- Team sticker applied where the referee can see it (R9). The sticker is also the inspection pass mark.
- No entanglement devices, adhesives, suction, magnets, liquids, loose material, weapons, sharp edges, glass, or sensor-jamming devices (R10 to R15). Paper test on every wheel (R11).
- Nothing designed to detach (R2).
- Battery commercially made, secured, and undamaged (R8). Wiring secured, no exposed conductors.
- Power cutoff demonstrated (R7).
- Manual start button and five-second delay demonstrated (R6).
- Robot shows its team number on the LEDs at power-on and responds to a test START and STOP from the inspection controller (R5, R9). Team confirms the robot never transmits and Bluetooth is off.